

Enterprise Google Cloud Data Platform Blueprint

Volume 01: Executive Architecture Guide

Executive strategy, architecture overview, decisions and roadmap

Document attribute	Value
Version	0.1 reference implementation
Date	2026-07-10
Intended audience	Executive, architecture, security, platform engineering, data, AI, SRE and FinOps stakeholders
Scope	Fortune 100 reference blueprint for petabyte-scale Google Cloud data platform
Status	Generated implementation starter set; verify service-specific settings against current Google Cloud documentation

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Inherited Fortune 100 Assumptions

Area	Assumption
Scale	10-50 PB historical data; 50-250 TB/day new data; billions of events/day
Organization	Global enterprise with central platform team and federated domain teams
Regulatory	GDPR, HIPAA, PCI DSS, SOC 2, ISO 27001, NIST and regional residency constraints may apply
Availability	Tier 0 RTO <1 hour and RPO <5 minutes; Tier 1 RTO <4 hours and RPO <15 minutes
Cloud posture	Hybrid and multi-cloud integration with Google Cloud as the enterprise data platform foundation
Security	Zero Trust, least privilege, private connectivity, service perimeters, CMEK for restricted data

1. Executive Summary

This volume defines the enterprise architecture vision for a Fortune 100 Google Cloud Data Platform. The platform standardizes ingestion, storage, processing, governance, analytics, AI, security, operations and cost management across global business units.

- Business outcomes include faster trusted analytics, governed AI enablement, reduced duplicated platforms, stronger privacy controls, reusable engineering patterns and transparent cost accountability.

2. Business Drivers and Requirements

The program is driven by legacy data-warehouse modernization, regulatory pressure, AI-readiness, faster product analytics, real-time decisioning and cost transparency. Requirements are prioritized using a must/should/could/future model.

Requirement	Priority	Architecture Response
Petabyte-scale analytics	Must	BigQuery curated serving layer with BigLake open lakehouse
Real-time and batch processing	Must	Pub/Sub, Dataflow, Datastream, Composer and Dataform
Governed AI	Should	Certified data products, Vertex AI, Vector Search and RAG guardrails
Regulated workloads	Must	VPC SC, IAM, KMS, DLP/Sensitive Data Protection and audit logging

Figure: Enterprise Google Cloud Data Platform - Conceptual Reference Architecture

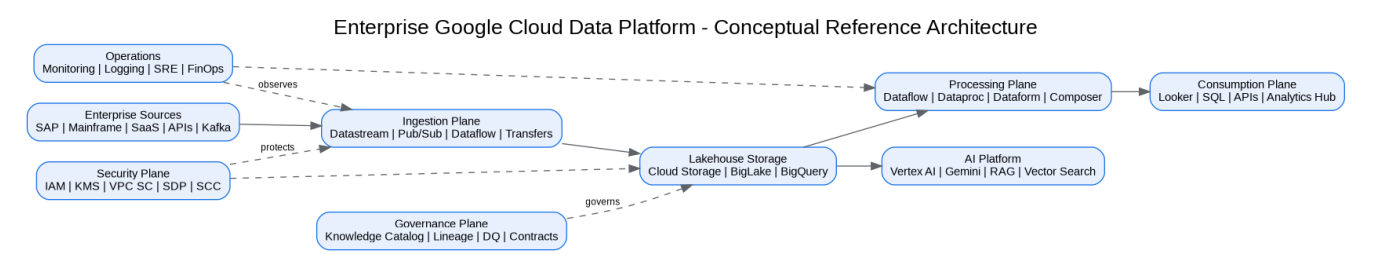
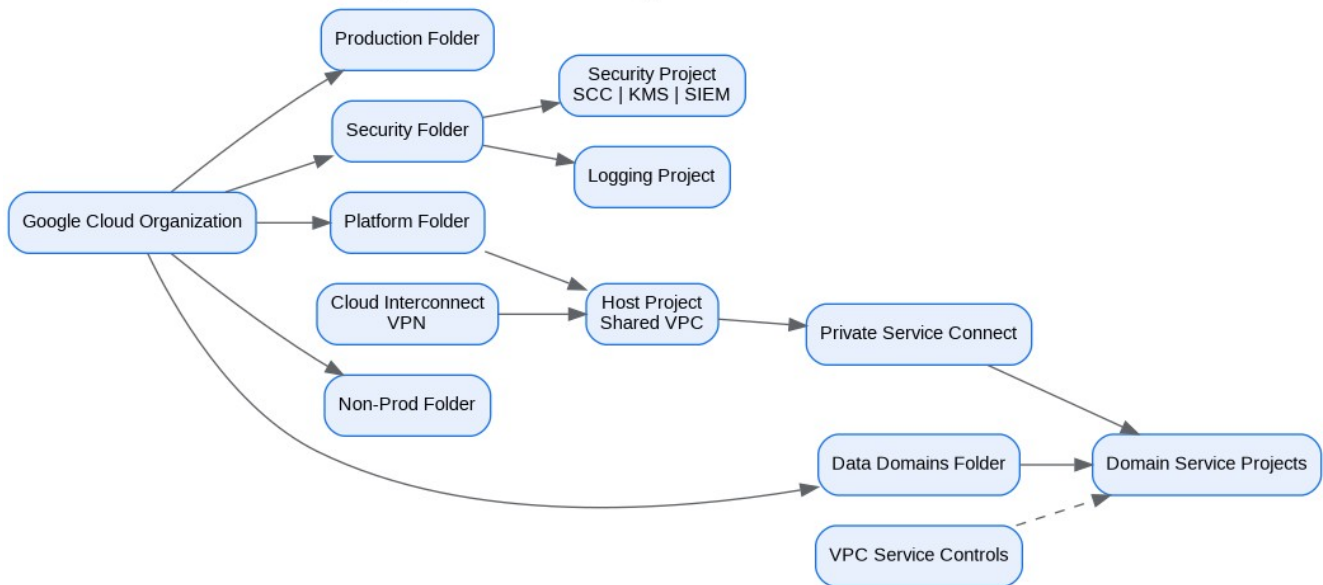


Figure: Google Cloud Landing Zone and Shared VPC

Google Cloud Landing Zone and Shared VPC



3. Architecture Principles

- Secure by design and privacy by design across every layer.
- Use managed and serverless services by default.
- Treat data as a product with ownership, quality and SLOs.
- Automate infrastructure, policies and deployment.
- Separate storage and compute; standardize metadata and observability.

4. Reference Architecture

The conceptual architecture separates source integration, ingestion, lakehouse storage, processing, governance, security, consumption, AI, operations and FinOps planes.

Service	Purpose	Benefits	Limitations / Pitfalls	Cost Controls
BigQuery	Enterprise analytics warehouse and curated serving layer	Serverless SQL, separation of storage/compute, strong ecosystem	Poor partitioning and unrestricted ad-hoc queries can create high costs	Reservations, labels, max bytes billed, partitioning, clustering
Cloud Storage	Raw, landing, archive and open-format lakehouse storage	Durable object storage, lifecycle policies, broad integration	Bucket sprawl and direct consumer access can weaken governance	Lifecycle policies, storage classes, retention controls
BigLake	Governed access to Cloud Storage data through BigQuery interfaces	Fine-grained access without direct bucket reads	Requires metadata and permission design	Table lifecycle, open-format governance
Dataflow	Batch and streaming pipelines using Apache Beam	Unified model, autoscaling, managed execution	Complex stateful pipelines require strong testing	Autoscaling, worker sizing, streaming engine
Pub/Sub	Event ingestion and decoupling	Reliable async messaging, retry and dead-letter support	Schema/version drift can impact consumers	Retention tuning and payload design
Knowledge Catalog	Metadata, glossary, lineage, governance and data discovery	Enterprise context layer for data and AI	Incomplete stewardship reduces trust	Metadata ownership and lifecycle controls

5. Roadmap

The transformation should progress from enterprise foundation to platform MVP, domain pilots, migration waves, AI enablement and continuous optimization.

Phase	Duration	Outcome
1. Foundation	8-12 weeks	Landing zone, security baseline, CI/CD, logging
2. MVP	12-16 weeks	Raw/curated layers, reference pipelines, catalog
3. Pilot domain	8-12 weeks	Customer or finance data product in production
4. Scale-out	6-18 months	Migration waves and domain onboarding

Architecture Decision Records

ADR	Decision	Problem	Alternatives	Decision	Tradeoffs/Risks	Mitigation
ADR-001	Google Cloud as enterprise data platform	Fragmented analytics estate	Continue legacy; multi-cloud split; Google Cloud data platform	Adopt Google Cloud data platform with hybrid integration	Requires operating-model change	Phased landing zone, domain pilots, and governance automation
ADR-002	BigQuery analytical core	Need PB-scale SQL analytics	Legacy EDW; Spark-only; third-party warehouse	Use BigQuery for curated/serving layers	Cost governance required	Reservations, labels, query controls, FinOps dashboards
ADR-003	Lakehouse with Cloud Storage and BigLake	Need open data and governed access	BigQuery-only; unmanaged object lake	Use Cloud Storage + BigLake + BigQuery	Open table governance maturity required	Catalog, policies, certified data-product lifecycle
ADR-004	Zero Trust and VPC SC	Exfiltration risk and regulated data	Network-only perimeter; IAM-only controls	Use identity, network, data and service perimeters	Service exceptions require testing	Perimeter dry runs and explicit ingress/egress policies

Risks and Mitigations

Risk	Impact	Mitigation
Underestimated legacy complexity	High	Complete source, pipeline, report and dependency assessment before migration waves
Data ownership gaps	High	Create domain owner RACI and data-product certification gate
Cost overrun	High	Enforce labels, budgets, reservations, query guardrails and monthly FinOps reviews
Security control drift	Critical	Deploy IaC, policy-as-code, automated IAM review and security posture monitoring
AI data leakage	Critical	Use certified datasets, service perimeters, DLP, RAG guardrails and model governance

Production Readiness Checklist

- All projects, networks, KMS keys and service accounts deployed through Terraform.
- All data products registered with owner, steward, classification, SLO, contract and support model.
- All restricted paths validated against VPC Service Controls and private connectivity requirements.
- All pipelines implement retries, dead-letter handling, reconciliation and operational metrics.
- All dashboards, alerts, runbooks and escalation routes tested before production cutover.
- All cost allocation labels present and billing export dashboards available to product owners.

References

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- Google Cloud Terraform blueprints and modules: <https://docs.cloud.google.com/docs/terraform/blueprints/terraform-blueprints>
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- Optimize costs with FinOps hub: <https://docs.cloud.google.com/billing/docs/how-to/finops-hub>
- VPC Service Controls supported products: <https://docs.cloud.google.com/vpc-service-controls/docs/supported-products>